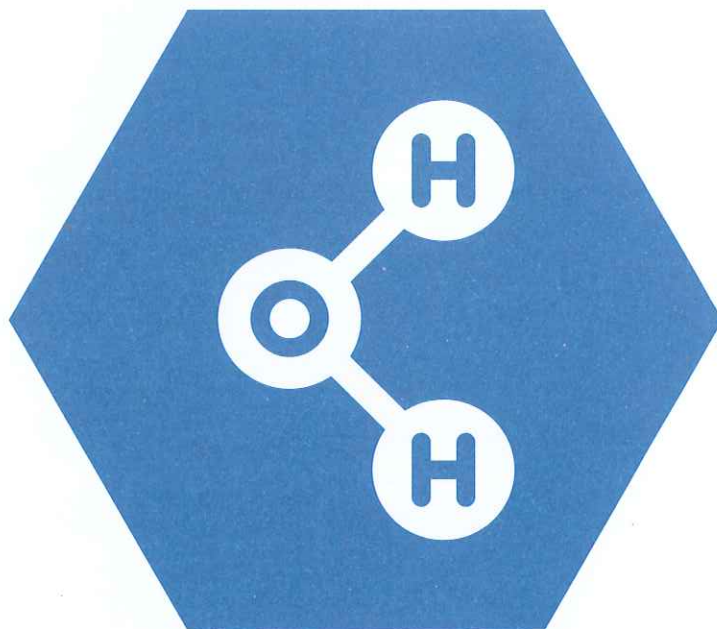




UNSW Global
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PAPER B



2014 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

HANNAH

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

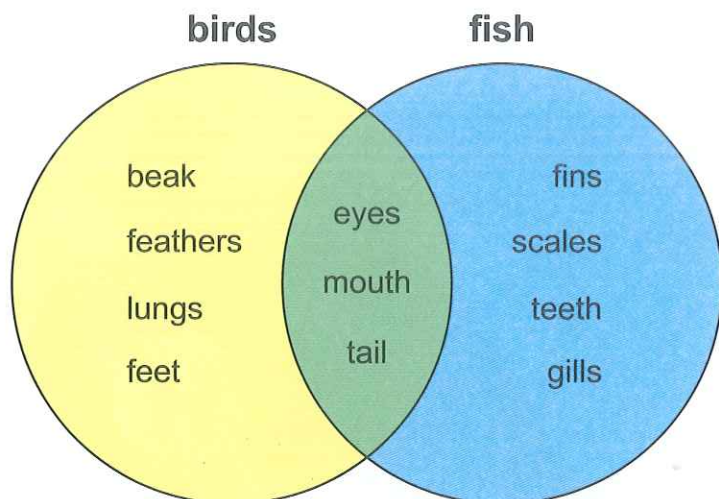
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. The diagram compares the features of birds with the features of fish.

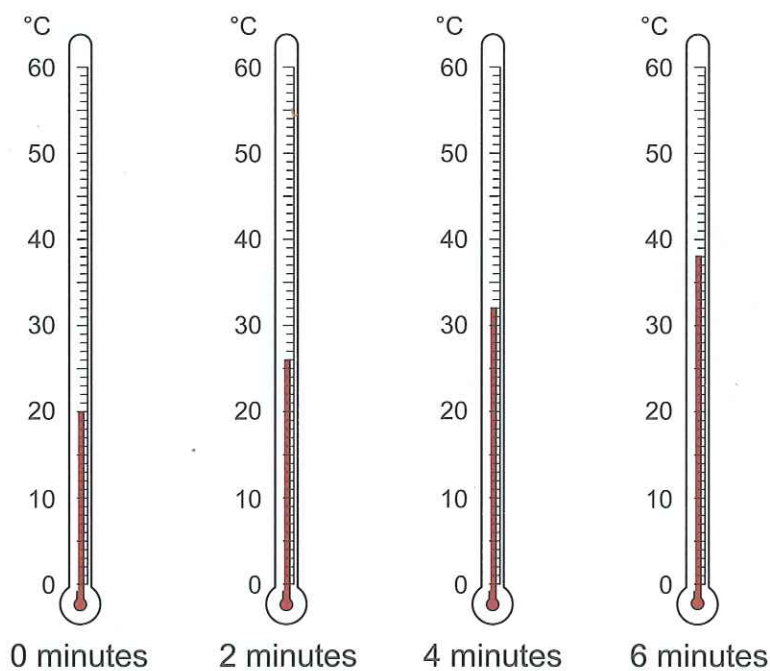


What are two features that birds and fish have in common?

- (A) mouth and tail
- (B) fins and teeth
- (C) teeth and eyes
- (D) feathers and lungs

2. Adam heated some water for seven minutes. He measured the rise in water temperature using a thermometer.

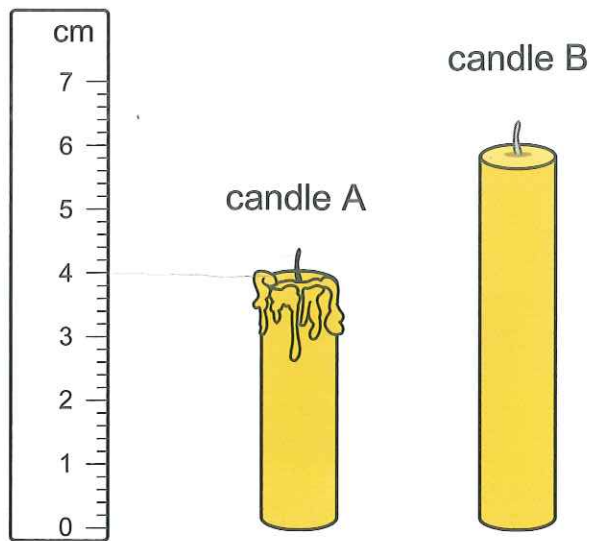
The thermometers show the temperature of the water every two minutes.



What is the water temperature likely to be at 7 minutes?

- (A) 38 °C
- (B) 40 °C
- (C) 41 °C
- (D) 44 °C

3. Milan had two identical candles. Milan lit candle A and let it burn for one hour. He then blew the flame out. Milan measured the height of the candles with a ruler.

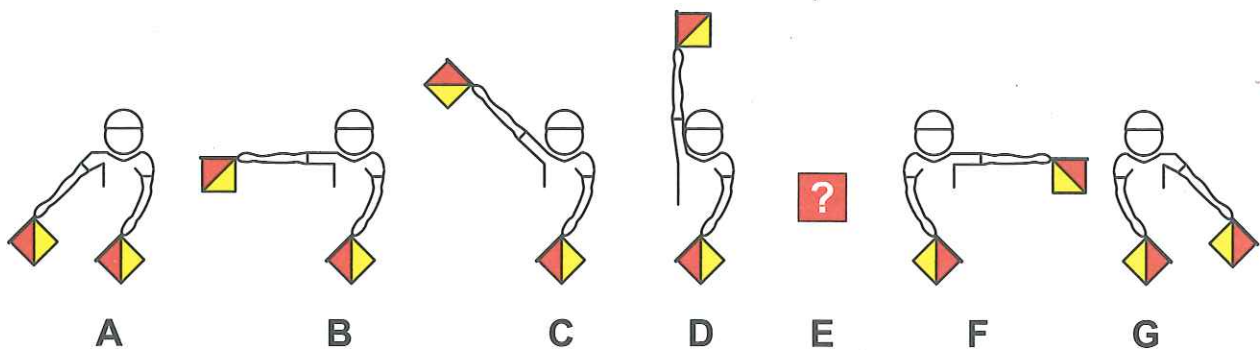


Milan then lit candle B.

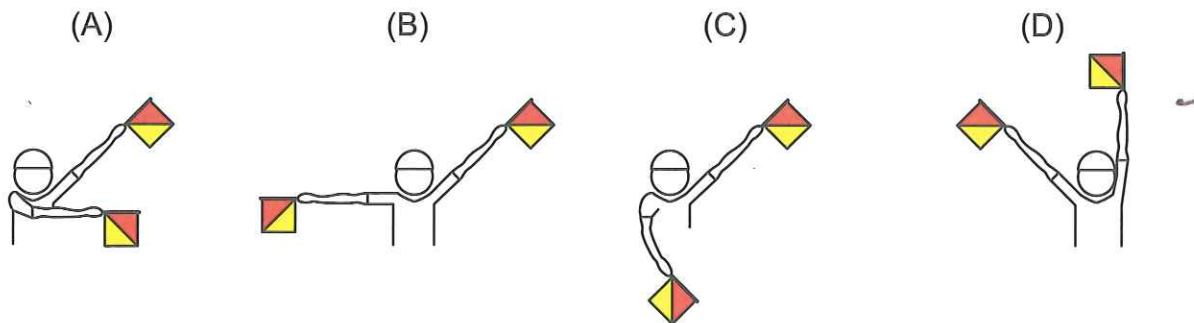
How long will it take for candle B to burn completely?

- (A) 2 hours
- (B) 3 hours
- (C) 4 hours
- (D) 6 hours

4. Flags can be used to send messages. The position of a pair of flags held by a person matches a letter of the alphabet. The positions of the flags follow a pattern.

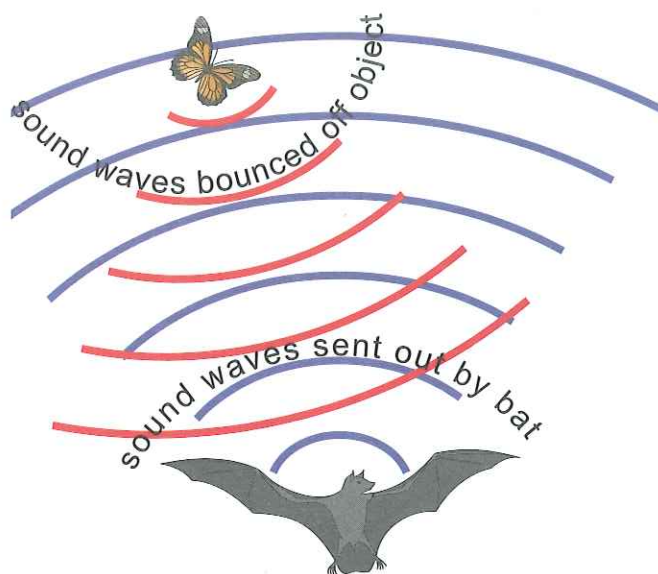


Which of the following represents the letter E?



5. Echoes are produced when sound waves bounce off solid objects. Bats use echoes to determine the distance between them and objects in their surroundings.

A bat sends out a series of high-pitched sounds. The times it took for sound waves to reach some objects and bounce back to the bat are recorded in the table.



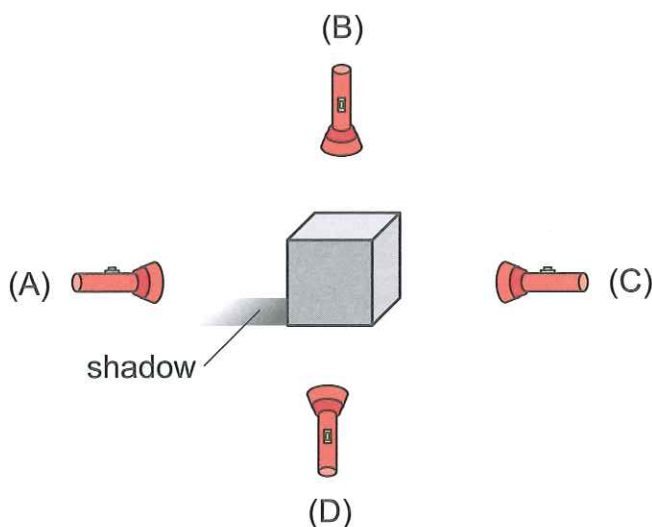
Object	Time taken for sound to reach object and bounce back to bat (seconds)
butterfly	0.25
tree	0.30
building	0.35
bird	0.20

Based on the data in the table, which conclusion is correct?

- (A) The bird is flying towards the bat.
- (B) The building is the furthest object from the bat.
- (C) The butterfly is the closest object to the bat.
- (D) The bird is further from the bat than the building.

6. A shadow is formed when an object blocks light.

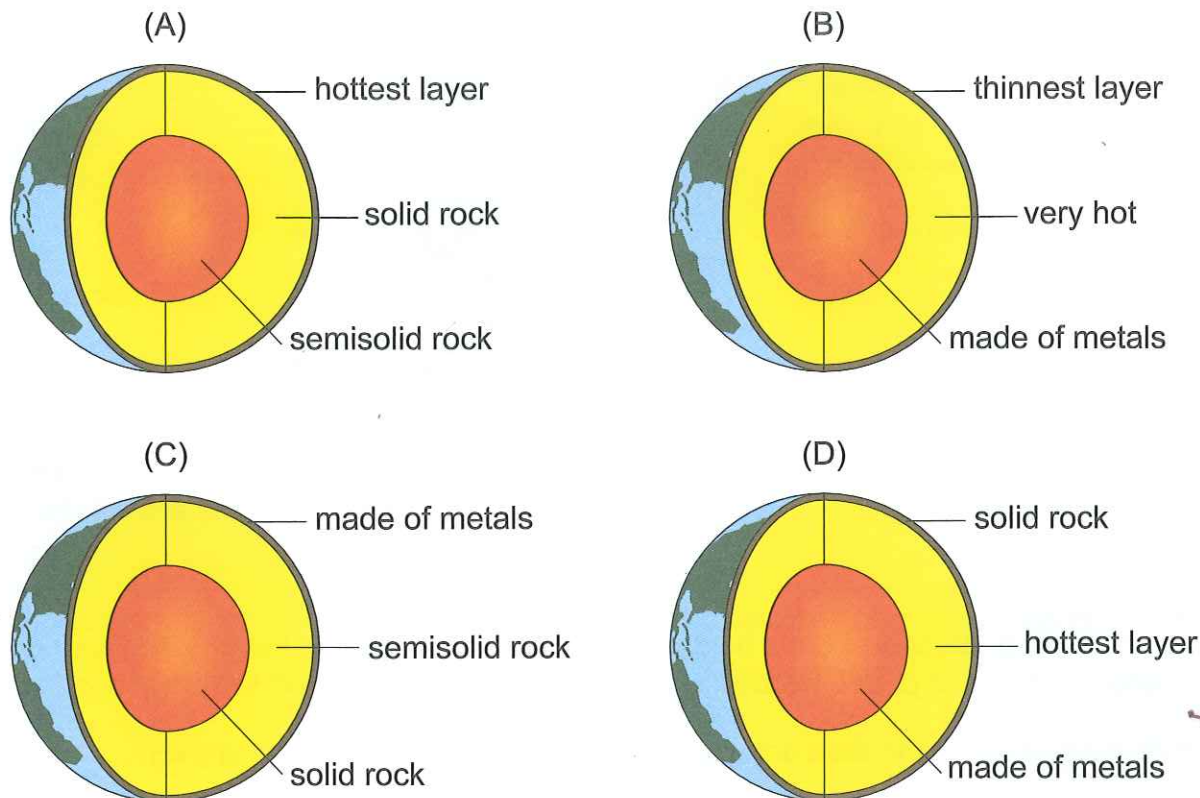
Which torch would make this shadow when it is switched on?



7. Earth is made up of three main layers: the crust, the mantle and the core.

- The crust is the thinnest layer and is solid rock. We live on the crust.
- The mantle is very hot and consists of mainly semisolid rock.
- The core is the deepest and hottest layer and is made of metals.

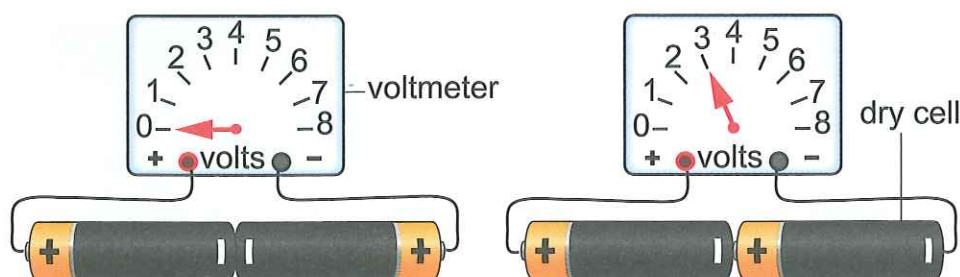
Which diagram correctly matches information about the layers of Earth?



8. A dry cell produces a voltage. This can be measured using a voltmeter.

When dry cells are joined, the total voltage depends on the number of cells and how they are connected.

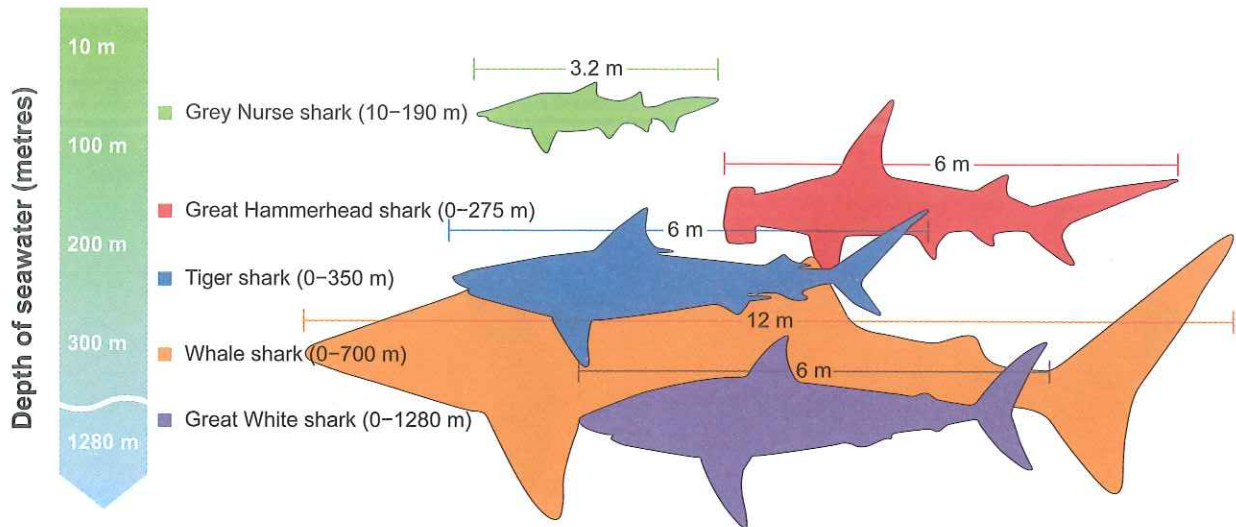
The diagrams show two ways that dry cells may be connected and the number of volts produced in each case.



How many volts does each dry cell produce?

- (A) 0 (B) $1\frac{1}{2}$ (C) 3 (D) $4\frac{1}{2}$

9. This diagram shows the name and size of different adult sharks and the depth of seawater at which each shark is found.

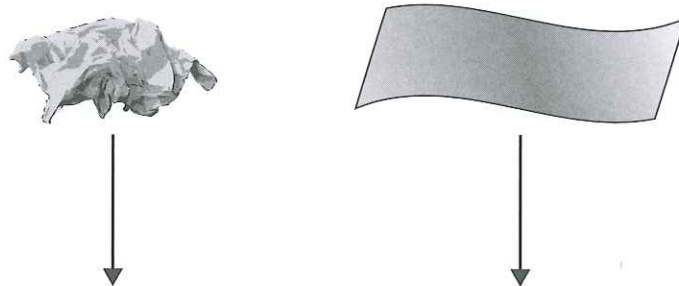


Which sharks would be found at depths between 300–750 m?

- (A) Grey Nurse and Tiger shark
(B) Great Hammerhead and Whale shark
(C) Tiger and Whale shark
(D) Great White, Tiger and Whale shark

10. When an object is dropped from a height, gravity causes it to fall to the ground. Sam took two identical sheets of paper. He crushed one sheet of paper and left the other one flat.

He took them outside and then dropped them from the same height as shown.

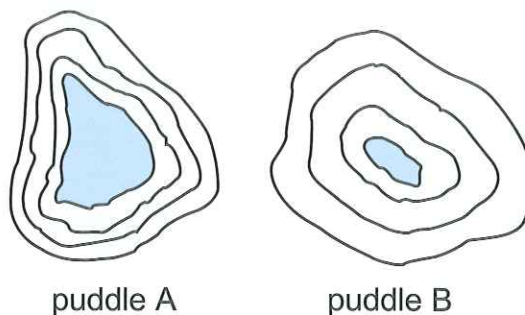


Which sheet of paper will hit the ground first and for what reason?

	Paper that will hit the ground first	Reason
(A)	crushed sheet	It is heavier.
(B)	flat sheet	There is more air pushing it back.
(C)	crushed sheet	There is less air pushing it back.
(D)	both hit the ground at the same time	Both have the same mass.

11. Sarah made two puddles on her floor using two different liquids. She used 100 mL of each liquid and poured them into separate puddles at the same time.

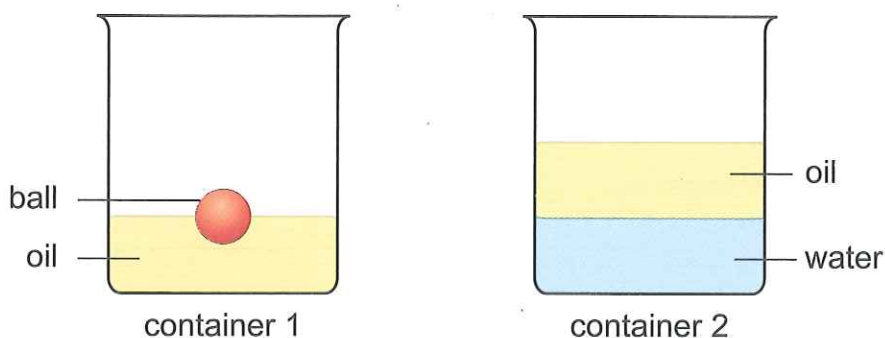
Sarah left the puddles to dry for 4 hours. Every hour she drew a line around what was left of each puddle as shown in the diagram.



What can Sarah conclude from her observations?

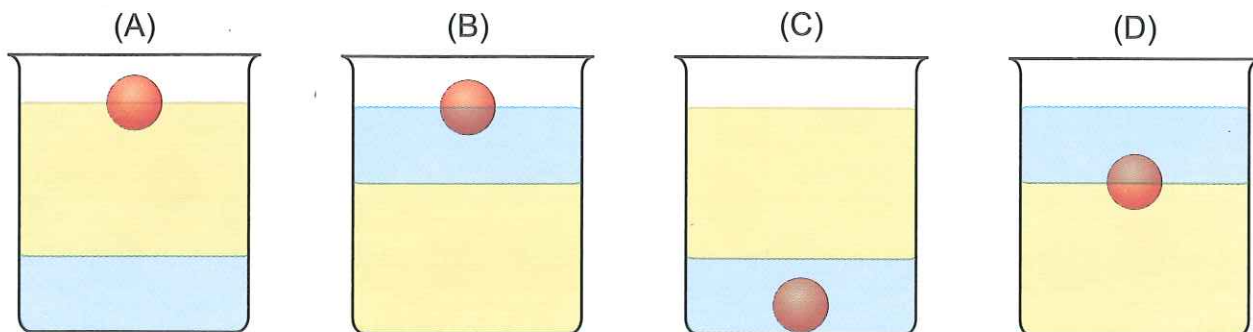
- (A) The liquid in puddle A was water.
- (B) Puddle B had more liquid than puddle A.
- (C) The liquid in puddle B stopped drying after 4 hours.
- (D) The liquid in puddle A dried more slowly than the liquid in puddle B.

12. Container 1 had oil and a small rubber ball in it. Container 2 had water and oil in it.



Jane poured the contents of container 2 into container 1.

Which diagram shows what container 1 looks like now?



For questions 13 and 14 use the information below.

The table shows the number of days it takes for different seeds to germinate when planted in soil at different temperatures. ^{grass}

Seeds	Days for seeds to germinate at different temperatures			
	10 °C	15 °C	20 °C	25 °C
onion	13	7	5	4
spinach	12	7	6	5
lettuce	7	4	3	2
cabbage	17	10	7	6
carrot	17	10	7	6
pea	14	9	8	6
tomato	43	14	8	6
corn	22	12	7	4
bean	–	30	17	6
cucumber	–	13	6	4

13. Which is the fastest germinating seed at any temperature?

- (A) spinach (B) tomato (C) cucumber (D) lettuce

14. Which seeds can be planted together and will germinate in the same time, at any time of the year?

- (A) onion and spinach (B) carrot and pea
(C) pea and tomato (D) cabbage and carrot

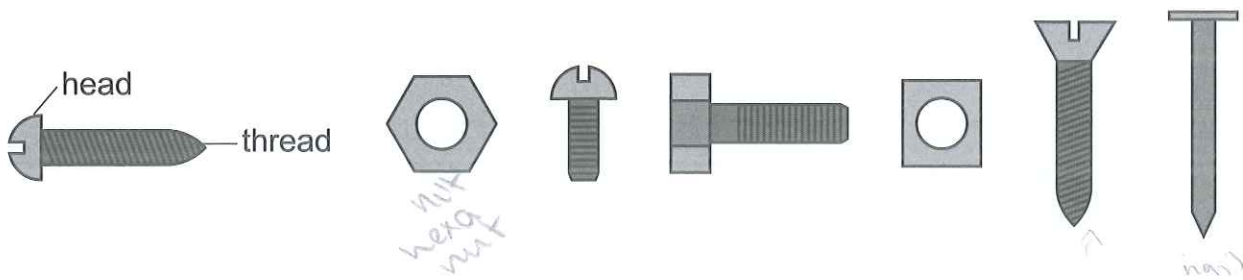
15. Sally and Jess performed some tests on a variety of rocks and entered their results into the table below.

Rock	Does it absorb water?	Does it split into layers?	Colour
slate	Yes	Yes	dark
marble	No	No	light
chalk	Yes	No	light
basalt	No	No	dark
pumice	Yes	No	light

Which rock does **NOT** absorb water, does **NOT** split into layers and is dark in colour?

- (A) marble (B) chalk (C) basalt (D) pumice

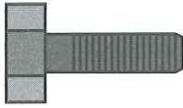
16. Jason found these objects in his toolbox.



He used this key to find the correct name for each object.

Question 1	Does it have a hole?	Yes	Go to Question 2.
		No	Go to Question 3.
Question 2	Does it have six sides?	Yes	It is a hexagonal nut.
		No	It is a square nut.
Question 3	Does it have a thread?	Yes	Go to Question 4.
		No	It is a nail.
Question 4	Does it have a rounded head?	Yes	It is a round-head screw.
		No	Go to Question 5.
Question 5	Does it have a pointy tip?	Yes	It is a countersunk screw.
		No	It is a hexagonal bolt.

What is the name of this object?

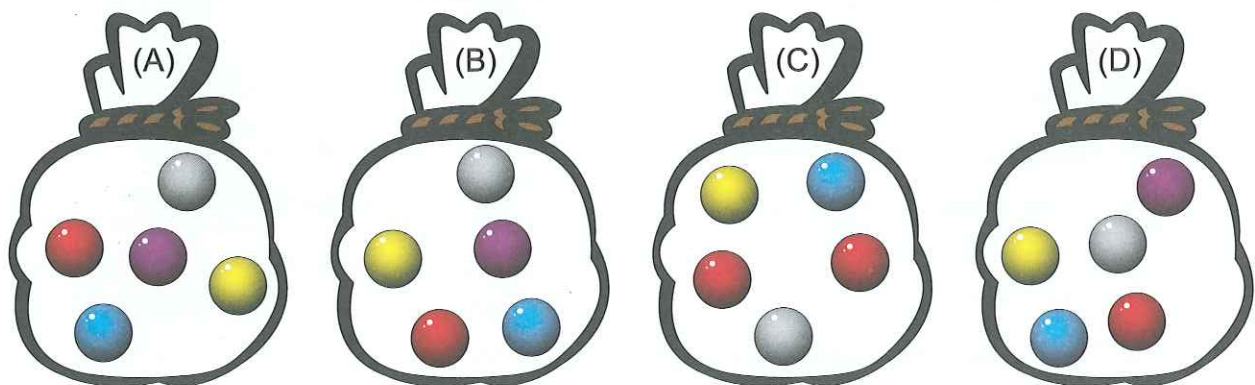


- (A) hexagonal nut
(C) countersunk screw

- (B) nail
(D) hexagonal bolt

17. Lisa put some coloured marbles into 4 bags labelled A, B, C and D.

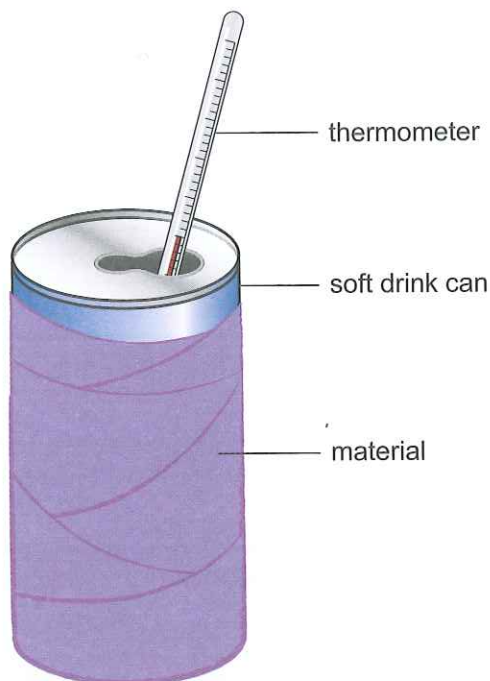
Which bag of marbles is different from the other three?



For questions 18 and 19 use the information below.

Amy did an experiment to test how well different materials reduce heat loss.

She wrapped three identical empty soft drink cans in a different material. She then put equal amounts of water at 50 °C into each can. After five minutes she measured the water temperature and recorded it in a table.



Material	Water temperature (°C) after five minutes
cotton	39
nylon	44
wool	42

18. Which of the following lists the materials from most effective at reducing heat loss to least effective?

- (A) nylon, wool, cotton
(C) wool, nylon, cotton

- (B) nylon, cotton, wool
(D) wool, cotton, nylon

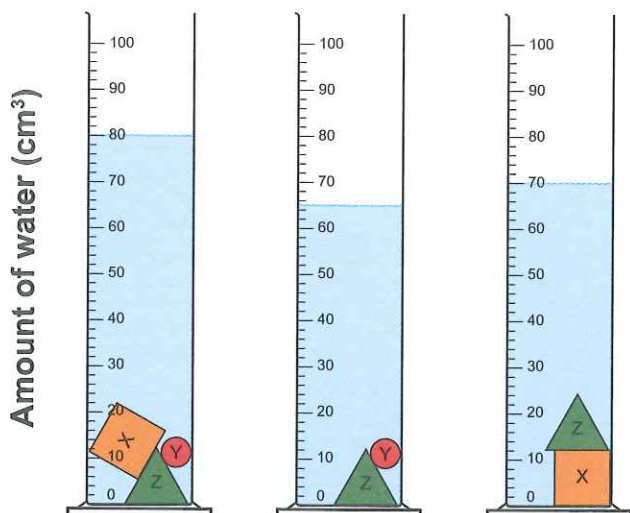
19. Amy wanted to use a control so she could compare her results.

Which of the following would be the best control for this experiment?

	Water amount	Water temperature	Covering materials
(A)	same as before	50 °C	all three together
(B)	same as before	50 °C	no material
(C)	same as before	room temperature	no material
(D)	no water	room temperature	no material

20. The volume of an object is equal to the volume of water it displaces.

Jason set up an investigation to find the volume of several objects using measuring cylinders and 50 mL of water in each measuring cylinder.



What is the volume of object Y?

- (A) 60 cm^3
(B) 15 cm^3
(C) 10 cm^3
(D) 5 cm^3

21. These photographs show what the Moon looked like on some nights during September. The photographs were taken from the same place on Earth, on the dates shown.



What would the Moon have looked like on September 26?



22. Ben has four test tubes containing sugar solutions of different concentrations. The sugar concentrations are 5%, 10%, 15% and 20%.

Ben knows the following:

- test tube 4 contains the highest concentration
- test tube 1 does not have the lowest concentration
- test tube 2 contains a greater sugar concentration than test tube 1.

Which test tube contains the 15% sugar solution?

- (A) 1 (B) 2 (C) 3 (D) 4

23. Matt was walking along the sand when he saw these footprints.



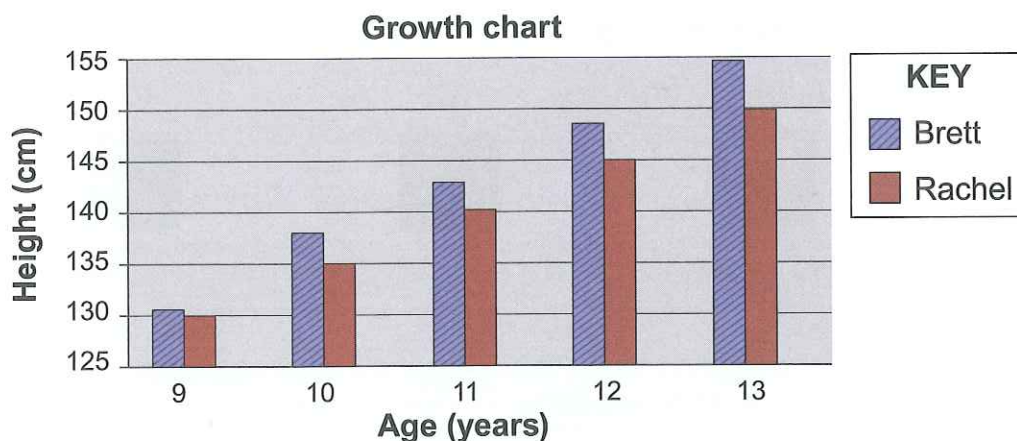
Which option shows the correct order of footprints on the sand, from first footprint to last footprint?

- (A) cat, horse, human, bird
(B) bird, human, horse, cat
(C) horse, human, bird, cat
(D) human, bird, horse, cat

For questions 24 and 25 use the information below.

Brett and Rachel had their heights measured on their birthdays over a number of years.

The growth chart shows this information.



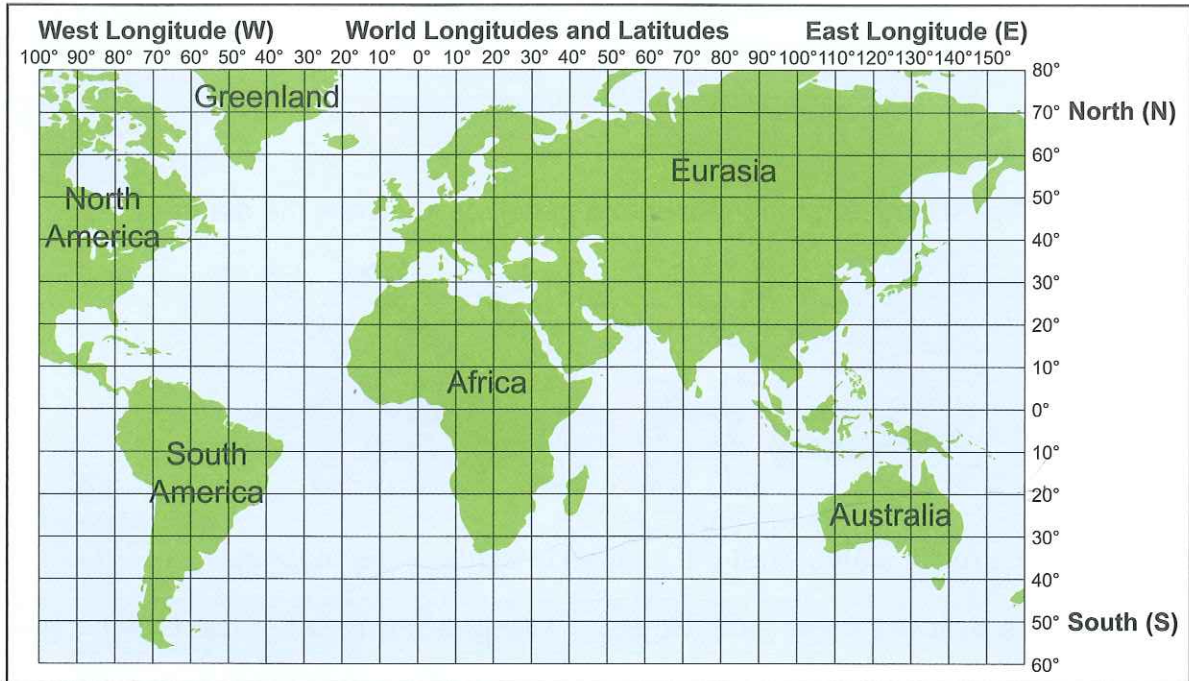
24. How many centimetres did Brett grow between his 10th birthday and his 11th birthday?

- (A) 5 (B) 7 (C) 23 (D) 143

25. What conclusion can be drawn from the information in the growth chart?

- (A) Brett is older than Rachel.
(B) Brett and Rachel have the same birthday.
(C) Brett was taller than Rachel on his 10th birthday.
(D) Brett grew faster than Rachel between the ages of 9 and 13.

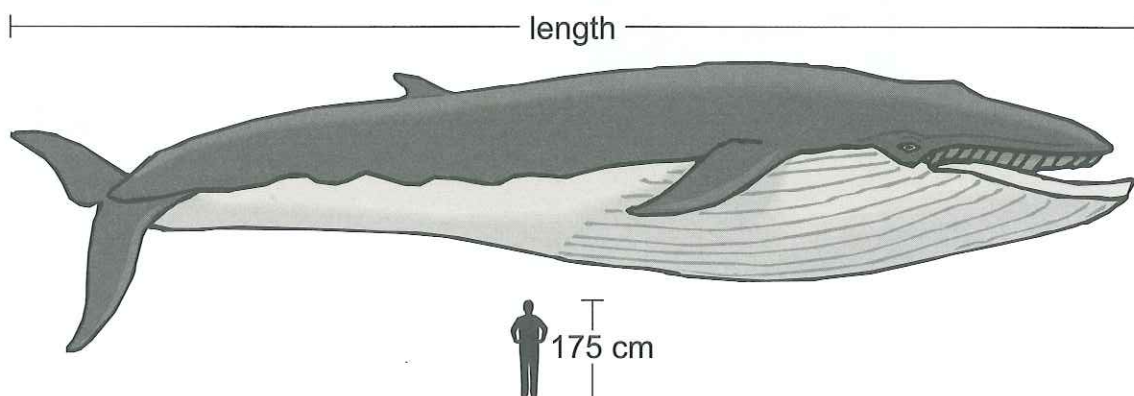
26. This is a picture of the fossil skull of *Cynognathus* that was found at location 20° S and 45° W.



Where was the fossil found?

- (A) Eurasia (B) Australia
(C) Africa (D) South America

27. The Blue Whale, a marine mammal, is the largest animal known to have lived on Earth.

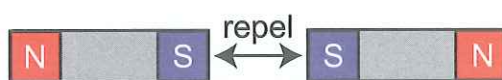


The man and the Blue Whale are drawn to the same scale.

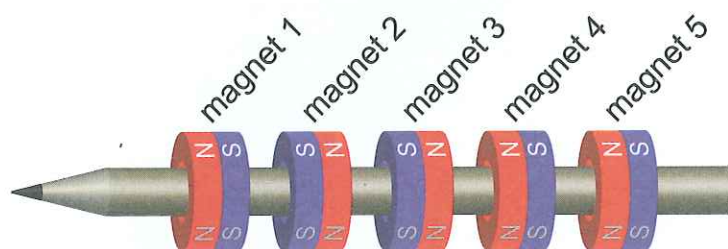
What is the approximate length of the Blue Whale?

- (A) 20 m (B) 30 m (C) 40 m (D) 50 m

28. Magnets have a north pole and a south pole. When magnets are placed near one another, they can either attract or repel one another as shown in the diagram.



Frans placed five strong ring magnets onto a pencil as shown in the diagram.

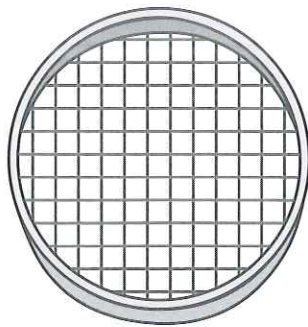


Which option correctly states what will happen to some of the magnets?

	Magnets that will repel each other	Magnets that will attract each other
(A)	magnet 1 and magnet 2	magnet 3 and magnet 4
(B)	magnet 3 and magnet 4	magnet 2 and magnet 3
(C)	magnet 4 and magnet 5	magnet 1 and magnet 2
(D)	magnet 2 and magnet 3	magnet 4 and magnet 5

For questions 29 and 30 use the information below.

A sieve is used to separate particles of different sizes. Toby has four sieves and four foods.



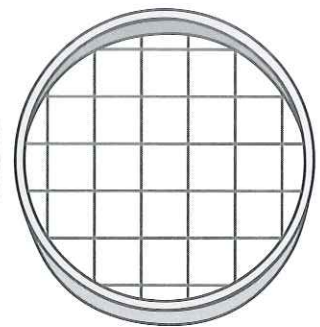
sieve W



sieve X



sieve Y



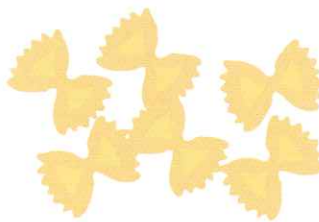
sieve Z



black-eye peas



rice



dried pasta



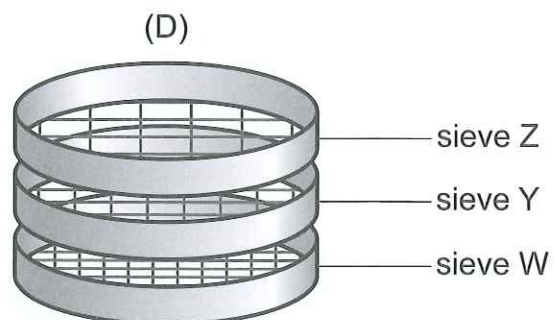
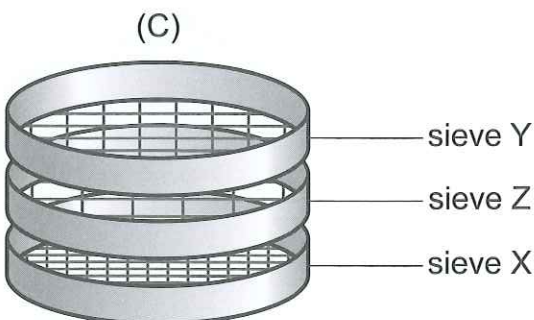
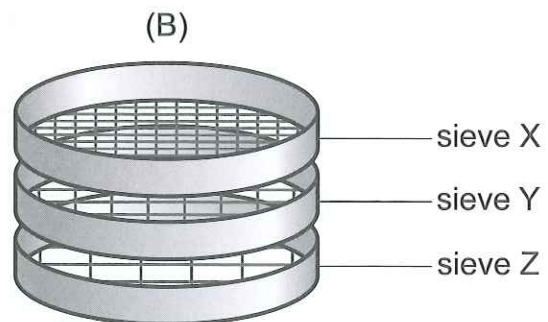
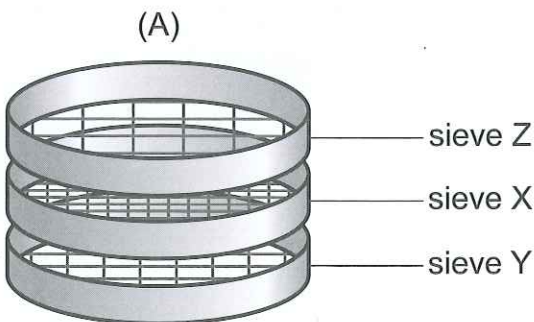
lentils

29. Which sieve will allow only two of these foods to pass through?

- (A) W (B) X (C) Y (D) Z

30. Toby only needs three sieves to separate the four foods. He places a mixture of the four foods into the top sieve and shakes the sieves.

Which combination of sieves will separate the four foods?



Acknowledgment

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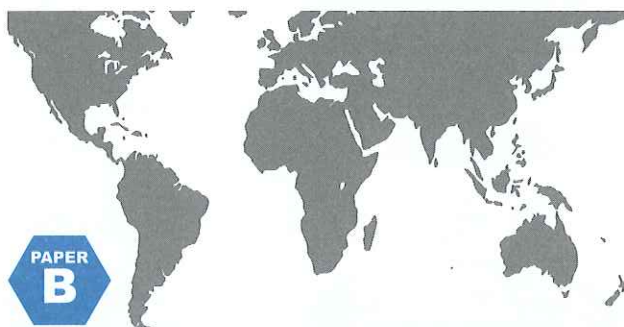


Sources

Question 26 Fossil skull of *Cynognathus* image adapted from a photograph by Ghedoghedo <<<http://en.wikipedia.org/wiki/File:Cynognathus.JPG>>> licensed under the Creative Commons Attribution-Share Alike 3.0 Unported licence.

The following year levels should sit THIS Paper:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific Region	Year 5
Singapore	Primary 3
South Africa	Grade 4



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- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

FIRST NAME										LAST NAME									
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☐	☐	☐	☐	☐	☐	☐	☐	☐	☐										
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B	B	B	B	B	B	B	B	B	B										
C	C	C	C	C	C	C	C	C	C										
D	D	D	D	D	D	D	D	D	D										
E	E	E	E	E	E	E	E	E	E										

FIRST NAME **LAST NAME**

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E	○	○	○

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●	A	A	A	A	A
B	B	B	●	B	B
C	C	C	C	C	○
D	D	D	D	D	○
E	E	E	E	E	○

FIRST NAME										LAST NAME				
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☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	☐ B	
☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	☐ C	
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☐ Male ☐ Female



☐ Yes ☐ No

Town / suburb: _____

Today's date: / / Postcode:

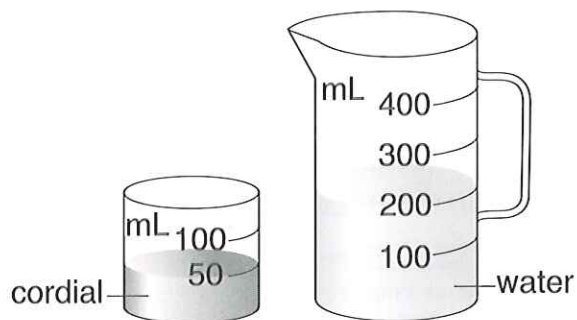
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(2)	(2)		(2)	(2)	(2)
(3)	(3)		(3)	(3)	(3)
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(5)			(5)	(5)	(5)
(6)			(6)	(6)	(6)
(7)			(7)	(7)	(7)
(8)			(8)	(8)	(8)
(9)			(9)	(9)	(9)

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval (D), as shown.

(A) (B) (C) (D)



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
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- 17 (A) (B) (C) (D)
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- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)

- 31 (A) (B) (C) (D)
- 32 (A) (B) (C) (D)
- 33 (A) (B) (C) (D)
- 34 (A) (B) (C) (D)
- 35 (A) (B) (C) (D)
- 36 (A) (B) (C) (D)
- 37 (A) (B) (C) (D)
- 38 (A) (B) (C) (D)
- 39 (A) (B) (C) (D)
- 40 (A) (B) (C) (D)



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2014 Science Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper B
1	A
2	C
3	B
4	C
5	B
6	C
7	B
8	B
9	D
10	C
11	D
12	A
13	D
14	D
15	C
16	D
17	C
18	A
19	B
20	C
21	D
22	B
23	B
24	A
25	D
26	D
27	A
28	B
29	A
30	D

(Please turn over)